

Linear Regression (multiple features with)

Ex: House Price Prediction

x_1	x_2	x_3	
Area (sqft)	Rooms	Locality Score	Price (\$)
500	2	8	800K
3200	8	7	1650K
900	4	10	1280K
1800	3	5	...
...	...	...	...

$m \times n$

y

Notation

$m \rightarrow$ no. of example

$n \rightarrow$ no. of features

$X \rightarrow$ input data matrix of shape $(m \times n)$

$y \rightarrow$ Target value

$X_{ij} \rightarrow$ j th feature of i th example

$\theta \rightarrow$ weights (parameters) of shape $(n+1) \times 1$

$\hat{y} \rightarrow$ hypothesis (outputs a real number)

Hypothesis (step 1)

$$\Rightarrow h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

$$\Rightarrow h_{\theta}(x) = \theta_0 + \sum_{i=1}^n \theta_i x_i$$

$$\Rightarrow h_{\theta}(x) = \theta_0 x_0 + \sum_{i=1}^n \theta_i x_i \quad \left[\text{where } x_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}_{m \times 1} \right]$$

$$\Rightarrow h_{\theta}(x) = \left(\sum_{i=0}^n \theta_i x_i \right) \quad \text{So, } \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_n \end{bmatrix}_{(n+1, 1)}$$

$$\Rightarrow \boxed{h_{\theta}(x) = \theta^T \cdot x}$$

$$\Rightarrow h_{\theta}(x) = [\theta_0 \ \theta_1 \ \theta_2 \ \dots \ \theta_n] \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\Rightarrow h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$$

Step-2 Loss function - Mean squared error

$$J = \frac{1}{2m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2$$

$$= \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$= \frac{1}{2m} \sum_{i=1}^m (\theta^T x^{(i)} - y^{(i)})^2$$

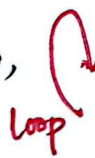
Step-3 Training & Gradient Update

→ Learn the parameters by reducing the loss using gradient descent algorithm.

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$\Rightarrow \frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{m} \cdot \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot (x_j^{(i)})$$

$$\Rightarrow \frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x_j^{(i)}$$

So,  $\theta_j = \theta_j - \eta \cdot \frac{\partial J(\theta)}{\partial \theta_j}$

Step-4

Let's coding

GitHub